Statistics and data managment

Dr Giulia Nicole Baldrighi

Turku PET Centre, Clinical Neurosciences

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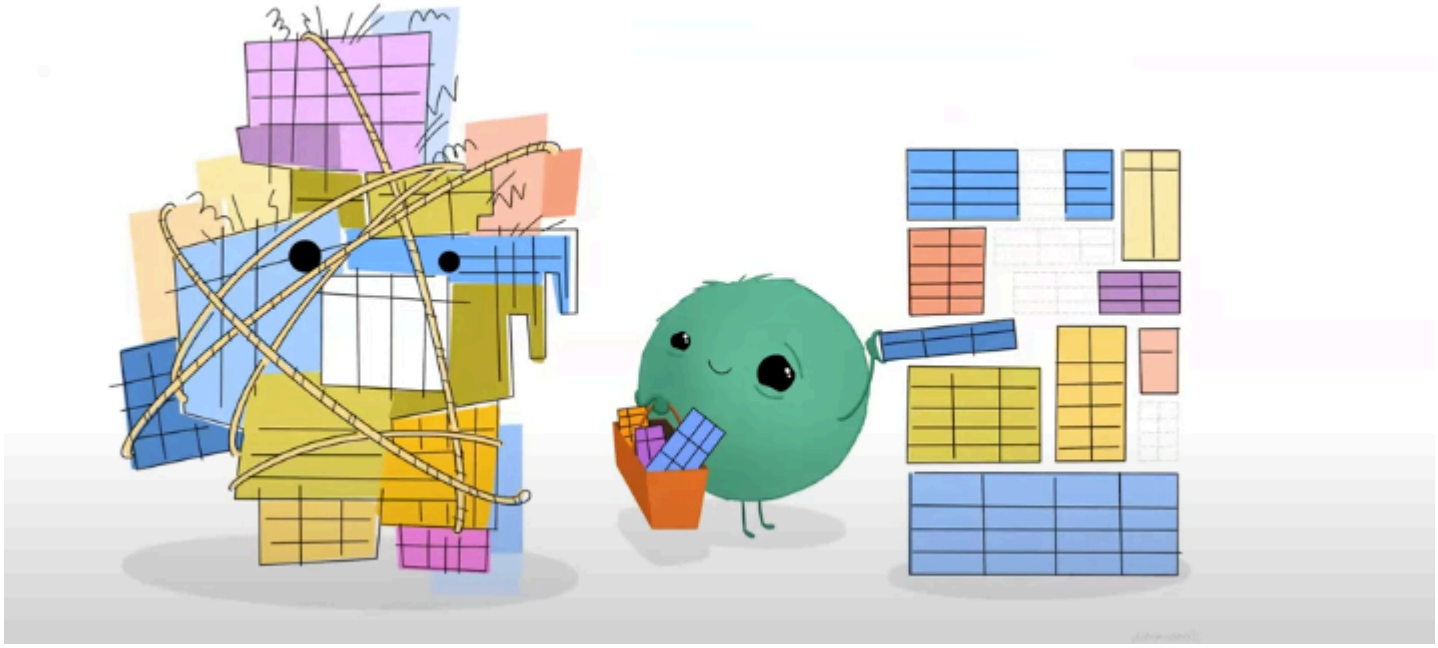


Let's first undertsand why statistics is important rather than focusing on what statistics literally means

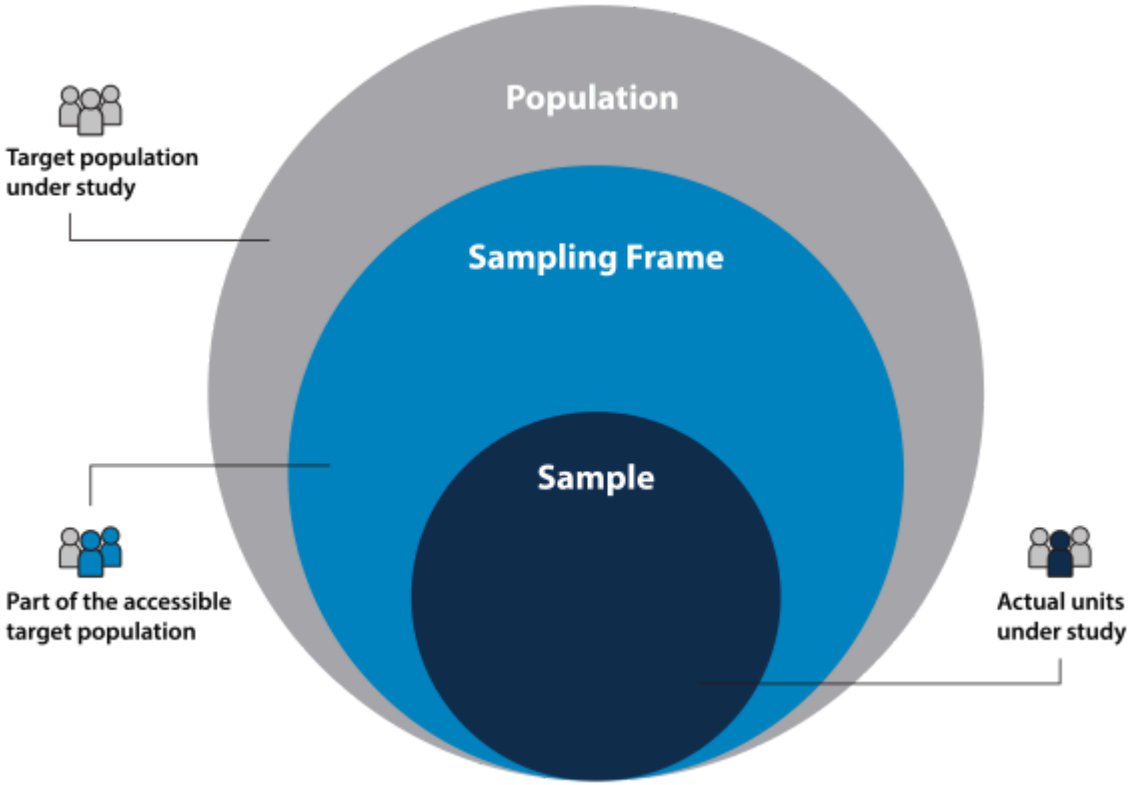
Purpose of statistics

- It helps researchers to **translate the information**
- Information **is the description of a phenomenon**
-
  - with its variability,
-
  - consequences
-
  - and interactions
- Information is not what the observer expects from science
- **Tip**: do not think to stress the data to obtain results

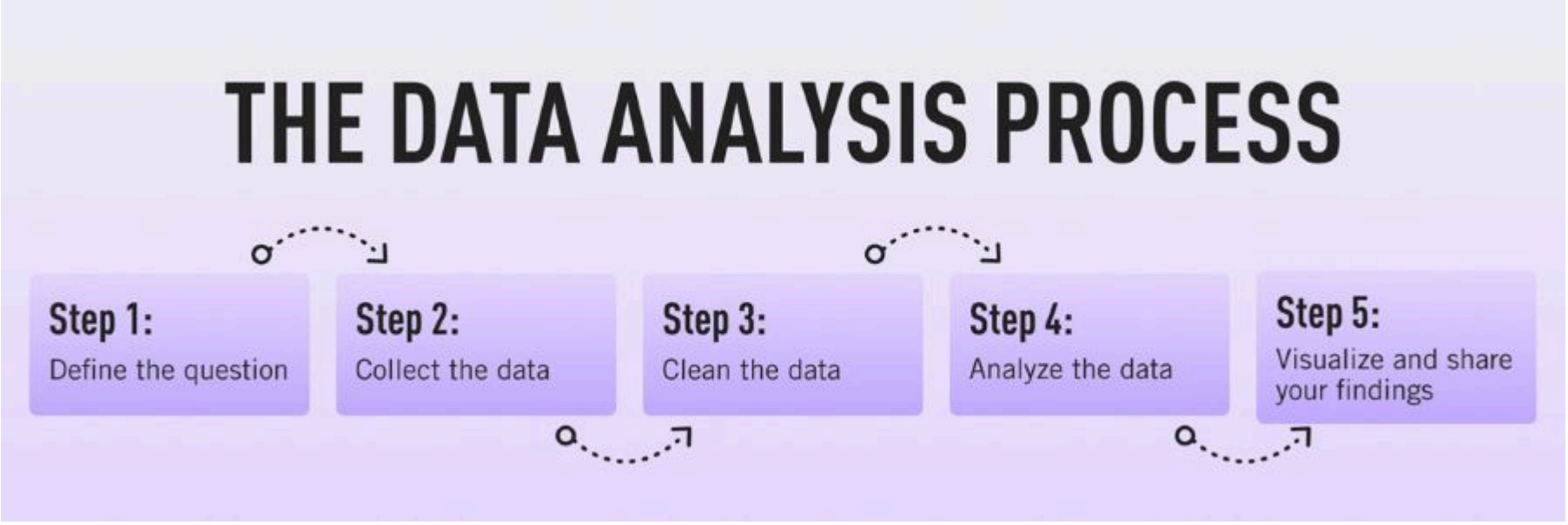
Good practice

- Think of a good question (if you have any)
- Collect data (reasonably) and follow good practice to analyse data and decode information
- Remember that there is not only one correct approach to approach data collection, analysis and interpretation, but there are several bad practice
- **Tip**: try to avoid them

Collecting data = codification of the information



Your personal dataset can contain more labels and information, but you must organize a proper dataset to follow statistical analyses



First aim is to build a functional dataset

Data structure

Variable types

Variable values

Missing data

Data structure

Cell

Columns

pat_id	treatment	start_date	start_bp_sys
001	upa	2022-01-04	90
002	uste	2022-03-15	101
003	oza	2022-04-04	98

Rows

UC Retrospective Study
Comparing Upa, Uste, Oza
6m Rx

pat_id	treatment	start_date	ethnic	race	dob	start_bp	pre/post_wt_kg	start_mes	start_bss	Start Abd score	start_sys
001	upa	44208	hispanic	Caucasian	1/7/2005	114/72	84/82	3	77	78	78
002	uste	44215	not hispanic	Caucasian	4/13/1937	132/86	77/77	2	53	51	57
003	oza	44230	not hispanic	African-American	6/6/1946	124/92	74/75	1	27	28	31
004	upa	44245	not hispanic	Caucasian	7/14/1963	144/83	66/65	3	77	79	82
005	oza	44255	not hispanic	Mixed	5/12/1978	122/78	55/56	3	77	80	80
006	uste	44259	not hispanic	Other	4/3/1992	121/80	111/110	2	51	52	54
007	uste	44264	not hispanic	Asian	8/22/1955	133/74	133/130	3	79	79	83
008	oza	44999	Hispanic	afromerican	9/11/1974	116/73	74/76	3	78	81	80
009	upa	44276	NOT hispani	Caucasian	11/14/1984	118/66	82/80	2	54	53	54
010	oza	44278	hispamnic	Caucasian	12/20/1972	122/78	85/87	3	76	81	82
011	upa	44297	not hispanic	Hawaiian	10/10/1994	126/82	88/88	3	80	79	78
012	uste	44308	not hispanic	Caucasian	11/17/1993	114/68	89/90	3	79	76	80
013	upa	44313	not hispanic	Caucasian	2/27/1948	118/73	85/78	3	79	77	83
014	oza	44318	hispanic	African-American	4/22/1966	106/59	84/84	3	78	77	79
015	uste	44324	not hispanic	H/API	8/11/1978	112/69	77/77	2	53	52	57
016	uste	44329	not hispanic	African-American	10/28/1998	114/76	78/79	3	77	80	82

Variable types

- **Numeric**: integer or continuous
- **Qualitative**: categorical (categorical variables can be ordered)
- **Binary**: (0,1) absence/presence disease; male/female (M,F)
- Binary variables can be recoded in (0,1) or (1,2) depending on the purpose
- Depending on variable type, different summary statistics, plots and downstream analyses should be chosen

Dates

- YYYY-MM-DD

If two times (using any of the date or date/time classes) are subtracted, R will return the results in the form of a time difference, which represents a difftime object.

For example, New York City experienced a major blackout on July 13, 1997, and another on August 14, 2003. To calculate the time interval between the two blackouts, we can simply subtract the two dates, using any of the classes that have been introduced:

```
b1 = ISOdate(1977,7,13)
b2 = ISOdate(2003,8,14)
b2 - b1
```

```
## Time difference of 9528 days
```

If an alternative unit of time was desired, the difftime function could be called, using the optional units= argument can be used with any of the following values: "auto", "secs", "mins", "hours", "days", or "weeks". So to see the difference between blackouts in terms of weeks, we can use:

```
difftime(b2,b1,units='weeks')
```

```
## Time difference of 1361.143 weeks
```

```
difftime(b2,b1,units='days')
```

```
## Time difference of 9528 days
```

Variable values

- Values should be explicit and not implicit
- If there is a blank cell that means missing value: avoid blank cells for other purpose
- If the actual value of a blank cell should be zero, then fill with a zero

Missing data

- Missing data can be informative (if patients drop)
- Missing data can be due to incorrect measurement (NA should be removed)
- Other NAs are generally excluded but reported in exploratory and summary statistics
- Pay attention to the percentage of missing (variables with high NA% should be removed, >30%)
- Subjects with high percentage of missing data should be removed (>50%)
- But consideration on percentages have to be analysed in each singular case

Codebook

- Compile a clean dataset
- Provide a **codebook** of variables with coding table
- Also provide **metadata** of patients (if some experimental assays are performed, e.g. genome sequencing, exome sequencing, or notes on familial history, treatments if they are tough to report in the main dataset)
- The same ID must be used

Codebook

variable	position	Variable_Label	units	Codes	type
casenum	1	case number, from 1 to 581			double
year	2	2 digit year, from 68 (1968) to 80 (1980)		68-80	double
month	3	month, from 1 (January) to 12 (December)		1-12	double
age	4	age in years	years		double
race	5	race, coded as 1 = Black, 2 = White		1-2	double
sex	6	gender, coded as 0 = male and 1 = female		0-1	double
blood_wbc	7	white blood cell count in blood	thousands of cells per cubic millimeter		double
blood_neut_pct	8	Percentage of white blood cells that are neutrophils, considered indicative of acute bacterial infection, in blood	percentage		double
blood_band_pct	9	Band cells, also known as immature neutrophils, considered indicative of acute bacterial infection, in blood, in percent of the total white blood cells	percentage		double
blood_gluc	10	Blood glucose	milligrams per deciliter		double
csf_gluc	11	CSF (cerebrospinal fluid) glucose	milligrams per deciliter		double
csf_prot	12	CSF (cerebrospinal fluid) protein	milligrams per deciliter		double
csf_rbc	13	CSF (cerebrospinal fluid) red blood cells, indicative of contamination with blood (a traumatic spinal tap), which can throw off other csf variables when high	thousands of cells per cubic millimeter		double